Seán Gorman

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PROJECTS

MolecuLead

Full stack application connecting a biological knowledge graph, four bio-databases and an LLM via GraphRAG — enabling knowledge-grounded Q&A for scientific research in drug discovery.

Boltz-RNA [Top 2%]

Fine-tuned a biomolecular foundation model for enhanced RNA structure prediction in a competition, scoring among the top 2% of participants.

LeashDTI

Graph Neural Networks for Drug-Target Interaction prediction — Kaggle-hosted competition.

EXPERIENCE

Machine Learning Engineer

2024 – present

DeepLife

LLMs & Agentic Systems

— Built end-to-end multi-agent systems for biological research workflows — combining LLM orchestration, RAG pipelines, and autonomous agents for transcriptomics metadata curation and scientific literature discovery.
— Responsible for MCP tool servers, skills, sandboxed environments, tracing, evals, and observability. Within the first month, obs & evals yielded a **90% increase in information retrieval accuracy** and a significant reduction in hallucinations.
— Fine-tuned small language models via reinforcement learning (GRPO) for Tool-Integrated Reasoning (TIR) on domain-specific biological tasks; created fine-tuned LLM assistants for product Q&A and user support.

Biomolecular Modelling

— Developed large perturbation models (LPM) using scRNA-seq to predict cellular responses to chemical and genetic perturbations.
— Trained LPMs on datasets with **100+ million cells** across multi-GPU clusters.
— Applied ML to decode gene regulatory networks and predict cell state transitions.

ML Implementation Engineer

2024 – 2025

Self Employed

— Assisted companies integrating AI into their workflows, implementing advanced data processing with LLMs.
— Built patent analysis software for Highway Ventures — reduced time spent analysing a patent portfolio from 40 to 4 hours: a **10× efficiency boost**.

Staff Research Associate

2019 – 2022

University of California, SF — Gould Lab

— Spearheaded technical work across two projects on kidney dysfunction and KO cell line creation/analysis.
— Trained 4 new staff associates and junior specialists in genotyping and sequence analysis.
— Applied techniques including CRISPR-Cas9 transfection, western blots, immunolabeling, cell culture, and MEF harvesting.

EDUCATION

MSc Bioinformatics & Computational Biology

University of Lisbon · 2022 – 2024

"Deep Learning for Discovery of Drug Binding Activities to Orphan Targets"

Advanced ML · Data Mining · Systems Biology · Quantitative Methods

BSc Neuroscience (Hons)

University College Dublin · 2015 – 2019

"Modulation of Hippocampal Synaptic Transmission by Cannabidiol"

Molecular Genetics · Neuropharmacology · Data Modelling

AWARDS

Health Universe Hackathon 2023 — 3rd Place

AuRA: LLM-powered Autonomous Research Agent using custom RAG and Google Scholar, built with LangChain and OpenAI.

Stanford RNA Folding Challenge Part 2

Silver Medalist in the Kaggle-hosted Stanford challenge.

SKILLS

Python · SQL · R · JavaScript · TypeScript · Svelte · PyTorch · TensorFlow · scikit-learn · LangChain · NLP · Computer Vision · Reinforcement Learning · Distributed Training · Deep Learning · AWS · GCP · Docker · Git · Linux · Bioinformatics · Genomics · NGS · CRISPR · Systems Biology · Biostatistics · Data Mining · Big Data Analytics

ACADEMIC CITATIONS

- Massoudi, D., **Gorman, S.**, et al. (2023). Deletion of the Unfolded Protein Response Transducer IRE1α Is Detrimental to Aging Photoreceptors and to ER Stress-Mediated Retinal Degeneration. *Investigative Ophthalmology & Visual Science*, 64(4).
- Branyan, K., Labelle-Dumais, C., …, **Gorman, S.**, & Gould, D. B. (2023). Elevated TGFβ signaling contributes to cerebral small vessel disease in mouse models of Gould syndrome. *Matrix Biology*, 115, 48–70.
- Massoudi, D., **Gorman, S.**, et al. (2021). The UPR transducer—IRE1α—is required for photoreceptor health and protection against retinal degeneration. *Investigative Ophthalmology & Visual Science*, 62(8).